National Aeronautics and
Space Administration





NASA SYSTEMS ENGINEERING HANDBOOK





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NASA SP-2016-6105 Rev2 supersedes SP-2007-6105 Rev 1 dated December, 2007.

Cover photos: *Top left:* In this photo, engineers led by researcher Greg Gatlin have sprayed fluorescent oil on a 5.8 percent scale model of a futuristic hybrid wing body during tests in the 14- by 22-Foot Subsonic Wind Tunnel at NASA's Langley Research Center in Hampton, VA. The oil helps researchers "see" the flow patterns when air passes over and around the model. (NASA Langley/ Preston Martin) *Top right:* Water impact test of a test version of the Orion spacecraft took place on August 24, 2016, at NASA Langley Research Center (NASA) *Bottom left:* two test mirror segments are placed onto the support structure that will hold them. (NASA/Chris Gunn) *Bottom right:* This self-portrait of NASA's Curiosity Mars rover shows the vehicle at the "Mojave" site, where its drill collected the mission's second taste of Mount Sharp. (NASA/JPL-Caltech/MSSS)

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Preface

Since the initial writing of NASA/SP-6105 in 1995 and the following revision (Rev 1) in 2007, systems engineering as a discipline at the National Aeronautics and Space Administration (NASA) has undergone rapid and continued evolution. Changes include using Model-Based Systems Engineering to improve development and delivery of products, and accommodating updates to NASA Procedural Requirements (NPR) 7123.1. Lessons learned on systems engineering were documented in reports such as those by the NASA Integrated Action Team (NIAT), the Columbia Accident Investigation Board (CAIB), and the follow-on Diaz Report. Other lessons learned were garnered from the robotic missions such as Genesis and the Mars Reconnaissance Orbiter as well as from mishaps from ground operations and the commercial space flight industry. Out of these reports came the NASA Office of the Chief Engineer (OCE) initiative to improve the overall Agency systems engineering infrastructure and capability for the efficient and effective engineering of NASA systems, to produce quality products, and to achieve mission success. This handbook update is a part of that OCE-sponsored Agency-wide systems engineering initiative.

In 1995, SP-6105 was initially published to bring the fundamental concepts and techniques of systems engineering to NASA personnel in a way that recognized the nature of NASA systems and the NASA environment. This revision (Rev 2) of SP-6105 maintains that original philosophy while updating the Agency's systems engineering body of knowledge, providing guidance for insight into current best Agency practices, and maintaining the alignment of the handbook with the Agency's systems engineering policy.

The update of this handbook continues the methodology of the previous revision: a top-down compatibility with higher level Agency policy and a bottom-up infusion of guidance from the NASA practitioners in the field. This approach provides the opportunity to obtain best practices from across NASA and bridge the information to the established NASA systems engineering processes and to communicate principles of good practice as well as alternative approaches rather than specify a particular way to accomplish a task. The result embodied in this handbook is a top-level implementation approach on the practice of systems engineering unique to NASA. Material used for updating this handbook has been drawn from many sources, including NPRs, Center systems engineering handbooks and processes, other Agency best practices, and external systems engineering textbooks and guides.

This handbook consists of six chapters: (1) an introduction, (2) a systems engineering fundamentals discussion, (3) the NASA program/project life cycles, (4) systems engineering processes to get from a concept to a design, (5) systems engineering processes to get from a design to a final product, and (6) crosscutting management processes in systems engineering. The chapters are supplemented by appendices that provide outlines, examples, and further information to illustrate topics in the chapters. The handbook makes extensive use of boxes and figures to define, refine, illustrate, and extend concepts in the chapters.

Finally, it should be noted that this handbook provides top-level guidance for good systems engineering practices; it is not intended in any way to be a directive.

NASA/SP-2016-6105 Rev2 supersedes SP-2007-6105 Rev 1 dated December, 2007.

1.0 Introduction

1.1 Purpose

This handbook is intended to provide general guidance and information on systems engineering that will be useful to the NASA community. It provides a generic description of Systems Engineering (SE) as it should be applied throughout NASA. A goal of the handbook is to increase awareness and consistency across the Agency and advance the practice of SE. This handbook provides perspectives relevant to NASA and data particular to NASA.

This handbook should be used as a companion for implementing NPR 7123.1, Systems Engineering Processes and Requirements, as well as the Center-specific handbooks and directives developed for implementing systems engineering at NASA. It provides a companion reference book for the various systems engineering-related training being offered under NASA's auspices.

1.2 Scope and Depth

This handbook describes systems engineering best practices that should be incorporated in the development and implementation of large and small NASA programs and projects. The engineering of NASA systems requires a systematic and disciplined set of processes that are applied recursively and iteratively for the design, development, operation, maintenance, and closeout of systems throughout the life cycle of the programs and projects. The scope of this handbook includes systems engineering functions regardless of whether they are performed by a manager or an engineer, in-house or by a contractor.

There are many Center-specific handbooks and directives as well as textbooks that can be consulted for in-depth tutorials. For guidance on systems engineering for information technology projects, refer to Office of Chief Information Officer *Information Technology Systems Engineering Handbook Version 2.0*. For guidance on entrance and exit criteria for milestone reviews of software projects, refer to *NASA-HDBK-2203, NASA Software Engineering Handbook*. A NASA systems engineer can also participate in the NASA Engineering Network (NEN) Systems Engineering Community of Practice, located at https://nen.nasa.gov/web/se. This Web site includes many resources useful to systems engineers, including document templates for many of the work products and milestone review presentations required by the NASA SE process.

This handbook is applicable to NASA space flight projects of all sizes and to research and development programs and projects. While all 17 processes are applicable to all projects, the amount of formality, depth of documentation, and timescales are varied as appropriate for the type, size, and complexity of the project. References to "documents" are intended to include not only paper or digital files but also models, graphics, drawings, or other appropriate forms that capture the intended information.

For a more in-depth discussion of the principles provided in this handbook, refer to the NASA Expanded Guidance for SE document which can be found at *https://nen.nasa.gov/web/se/doc-repository*. This handbook is an abridged version of that reference.

TABLE 2.2-1 Project Life Cycle Phases

Phase		Purpose	Typical Outcomes
Pre-Formulation	**Pre-Phase A** Concept Studies	To produce a broad spectrum of ideas and alternatives for missions from which new programs/projects can be selected. Determine feasibility of desired system, develop mission concepts, draft system-level requirements, assess performance, cost, and schedule feasibility; identify potential technology needs, and scope.	Feasible system concepts in the form of simulations, analysis, study reports, models, and mock-ups
Formulation	**Phase A** Concept and Technology Development	To determine the feasibility and desirability of a suggested new system and establish an initial baseline compatibility with NASA's strategic plans. Develop final mission concept, system-level requirements, needed system technology developments, and program/project technical management plans.	System concept definition in the form of simulations, analysis, engineering models and mock-ups, and trade study definition
	Phase B Preliminary Design and Technology Completion	To define the project in enough detail to establish an initial baseline capable of meeting mission needs. Develop system structure end product (and enabling product) requirements and generate a preliminary design for each system structure end product.	End products in the form of mock-ups, trade study results, specification and interface documents, and prototypes
Implementation	**Phase C** Final Design and Fabrication	To complete the detailed design of the system (and its associated subsystems, including its operations systems), fabricate hardware, and code software. Generate final designs for each system structure end product.	End product detailed designs, end product component fabrication, and software development
	Phase D System Assembly, Integration and Test, Launch	To assemble and integrate the system (hardware, software, and humans), meanwhile developing confidence that it is able to meet the system requirements. Launch and prepare for operations. Perform system end product implementation, assembly, integration and test, and transition to use.	Operations-ready system end product with supporting related enabling products
	Phase E Operations and Sustainment	To conduct the mission and meet the initially identified need and maintain support for that need. Implement the mission operations plan.	Desired system
	Phase F Closeout	To implement the systems decommissioning/disposal plan developed in Phase E and perform analyses of the returned data and any returned samples.	Product closeout

processes (steps 1 through 9) in each project phase. The SE engine cycles five times from Pre-Phase A through Phase D. Note that NASA's management has structured Phases C and D to "split" the technical development processes in half in Phases C and D to ensure closer management control. The engine is bound by a dashed line in Phases C and D. Once a project enters into its operational state (Phase E) and closes out (Phase F), the technical work shifts to activities commensurate with these last two project phases. The next major horizontal band shows the eight technical management processes (steps 10 through 17) in each project phase. The SE engine cycles the technical management processes seven times from Pre-Phase A through Phase F.

Technology Maturity Assessment: A process to determine a system's technological maturity based on Technology Readiness Levels (TRLs).

Technology Readiness Level: Provides a scale against which to measure the maturity of a technology. TRLs range from 1, basic technology research, to 9, systems test, launch, and operations. Typically, a TRL of 6 (i.e., technology demonstrated in a relevant environment) is required for a technology to be integrated into an SE process.

Test: The use of a realized end product to obtain detailed data to verify or validate performance or to provide sufficient information to verify or validate performance through further analysis.

Test Readiness Review: A review that ensures that the test article (hardware/software), test facility, support personnel, and test procedures are ready for testing and data acquisition, reduction, and control.

Threshold Requirements: A minimum acceptable set of technical and project requirements; the set could represent the descope position of the project.

Tightly Coupled Programs: Programs with multiple projects that execute portions of a mission(s). No single project is capable of implementing a complete mission. Typically, multiple NASA Centers contribute to the program. Individual projects may be managed at different Centers. The program may also include contributions from other agencies or international partners.

Traceability: A discernible association among two or more logical entities such as requirements, system elements, verifications, or tasks.

Trade Study: A means of evaluating system designs by devising alternative means to meet functional requirements, evaluating these alternatives in terms of the measures of effectiveness and system cost, ranking the alternatives according to appropriate selection criteria, dropping less promising alternatives, and proceeding to the next level of resolution, if needed.

Trade Study Report: A report written to document a trade study. It should include: the system under analysis; system goals, objectives (or requirements, as appropriate to the level of resolution), and constraints; measures and measurement methods (models) used; all data sources used; the alternatives chosen for analysis; computational results, including uncertainty ranges and sensitivity analyses performed; the selection rule used; and the recommended alternative.

Trade Tree: A representation of trade study alternatives in which each layer represents some system aspect that needs to be treated in a trade study to determine the best alternative.

Transition: The act of delivery or moving of a product from one location to another. This act can include packaging, handling, storing, moving, transporting, installing, and sustainment activities.

Uncoupled Programs: Programs implemented under a broad theme and/or a common program implementation concept, such as providing frequent flight opportunities for cost-capped projects selected through AO or NASA Research Announcements. Each such project is independent of the other projects within the program.

Utility: A measure of the relative value gained from an alternative. The theoretical unit of measurement for utility is the "util."

Validated Requirements: A set of requirements that are well formed (clear and unambiguous), complete (agree with customer and stakeholder needs and

Appendix G: Technology Assessment/Insertion

G.1 Introduction, Purpose, and Scope

In 2014, the Headquarters Office of Chief Engineer and Office of Chief Technologist conducted an Agency-wide study on Technical Readiness Level (TRL) usage and Technology Readiness Assessment (TRA) implementation. Numerous findings, observations, and recommendations were identified, as was a wealth of new guidance, best practices, and clarifications on how to interpret TRL and perform TRAs. These are presently being collected into a NASA TRA Handbook (in work), which will replace this appendix. In the interim, contact HQ/Steven Hirshorn on any specific questions on interpretation and application of TRL/TRA. Although the information contained in this appendix may change, it does provide some information until the TRA Handbook can be completed.

Agency programs and projects frequently require the development and infusion of new technological advances to meet mission goals, objectives, and resulting requirements. Sometimes the new technological advancement being infused is actually a heritage system that is being incorporated into a different architecture and operated in a different environment from that for which it was originally designed. It is important to recognize that the adaptation of heritage systems frequently requires technological advancement. Failure to account for this requirement can result in key steps of the development process being given short shrift—often to the detriment of the program/project. In both contexts of technological advancement (new and adapted heritage), infusion is a complex process that is often dealt with in an ad hoc manner differing greatly from project to project with varying degrees of success.

Technology infusion frequently results in schedule slips, cost overruns, and occasionally even in cancellations or failures. In post mortem, the root cause of such events is often attributed to "inadequate definition of requirements." If such is indeed the root cause, then correcting the situation is simply a matter of defining better requirements, but this may not be the case—at least not totally.

In fact, there are many contributors to schedule slip, cost overrun, and project cancellation and failure—among them lack of adequate requirements definition. The case can be made that most of these contributors are related to the degree of uncertainty at the outset of the project and that a dominant factor in the degree of uncertainty is the lack of understanding of the maturity of the technology required to bring the project to fruition and a concomitant lack of understanding of the cost and schedule reserves required to advance the technology from its present state to a point where it can be qualified and successfully infused with a high degree of confidence. Although this uncertainty cannot be eliminated, it can be substantially reduced through the early application of good systems engineering practices focused on understanding the technological requirements; the maturity of the required technology; and the technological advancement required to meet program/project goals, objectives, and requirements.

TABLE G.1-1 Products Provided by the TA as a Function of Program/Project Phase

Gate	Product
KDP A: Transition from Pre-Phase A to Phase A	Requires an assessment of potential technology needs versus current and planned technology readiness levels, as well as potential opportunities to use commercial, academic, and other government agency sources of technology. Included as part of the draft integrated baseline. Technology Development Plan is baselined that identifies technologies to be developed, heritage systems to be modified, alternative paths to be pursued, fallback positions and corresponding performance descopes, milestones, metrics, and key decision points. Initial Technology Readiness Assessment (TRA) is available.
KDP B: Transition from Phase A to Phase B	Technology Development Plan and Technology Readiness Assessment (TRA) are updated. Incorporated in the preliminary project plan.
KDP C: Transition from Phase B to Phase C/D	Requires a TRAR demonstrating that all systems, subsystems, and components have achieved a level of technological maturity with demonstrated evidence of qualification in a relevant environment.

Source: NPR 7120.5.

A number of processes can be used to develop the appropriate level of understanding required for successful technology insertion. The intent of this appendix is to describe a systematic process that can be used as an example of how to apply standard systems engineering practices to perform a comprehensive Technology Assessment (TA). The TA comprises two parts, a Technology Maturity Assessment (TMA) and an Advancement Degree of Difficulty Assessment (AD^2). The process begins with the TMA which is used to determine technological maturity via NASA's Technology Readiness Level (TRL) scale. It then proceeds to develop an understanding of what is required to advance the level of maturity through the AD^2. It is necessary to conduct TAs at various stages throughout a program/project to provide the Key Decision Point (KDP) products required for transition between phases. (See TABLE G.1-1.)

The initial TMA provides the baseline maturity of the system's required technologies at program/project outset and allows monitoring progress throughout development. The final TMA is performed just prior to the Preliminary Design Review (PDR). It forms the basis for the Technology Readiness Assessment Report (TRAR), which documents the maturity of the technological advancement required by the systems, subsystems, and components demonstrated through test and analysis. The initial AD^2 provides the material necessary to develop preliminary cost and to schedule plans and preliminary risk assessments. In subsequent assessments, the information is used to build the Technology Development Plan and in the process, identify alternative paths, fallback positions, and performance descope options. The information is also vital to preparing milestones and metrics for subsequent Earned Value Management (EVM).

The TMA is performed against the hierarchical breakdown of the hardware and software products of the program/project PBS to achieve a systematic, overall understanding at the system, subsystem, and component levels. (See FIGURE G.1-1.)

G.2 Inputs/Entry Criteria

It is extremely important that a TA process be defined at the beginning of the program/project and that it be performed at the earliest possible stage (concept development) and throughout the program/project through PDR. Inputs to the process will vary in level of detail according to the phase of the program/project, and even though there is a lack of detail in Pre-Phase A, the TA will drive out the major critical

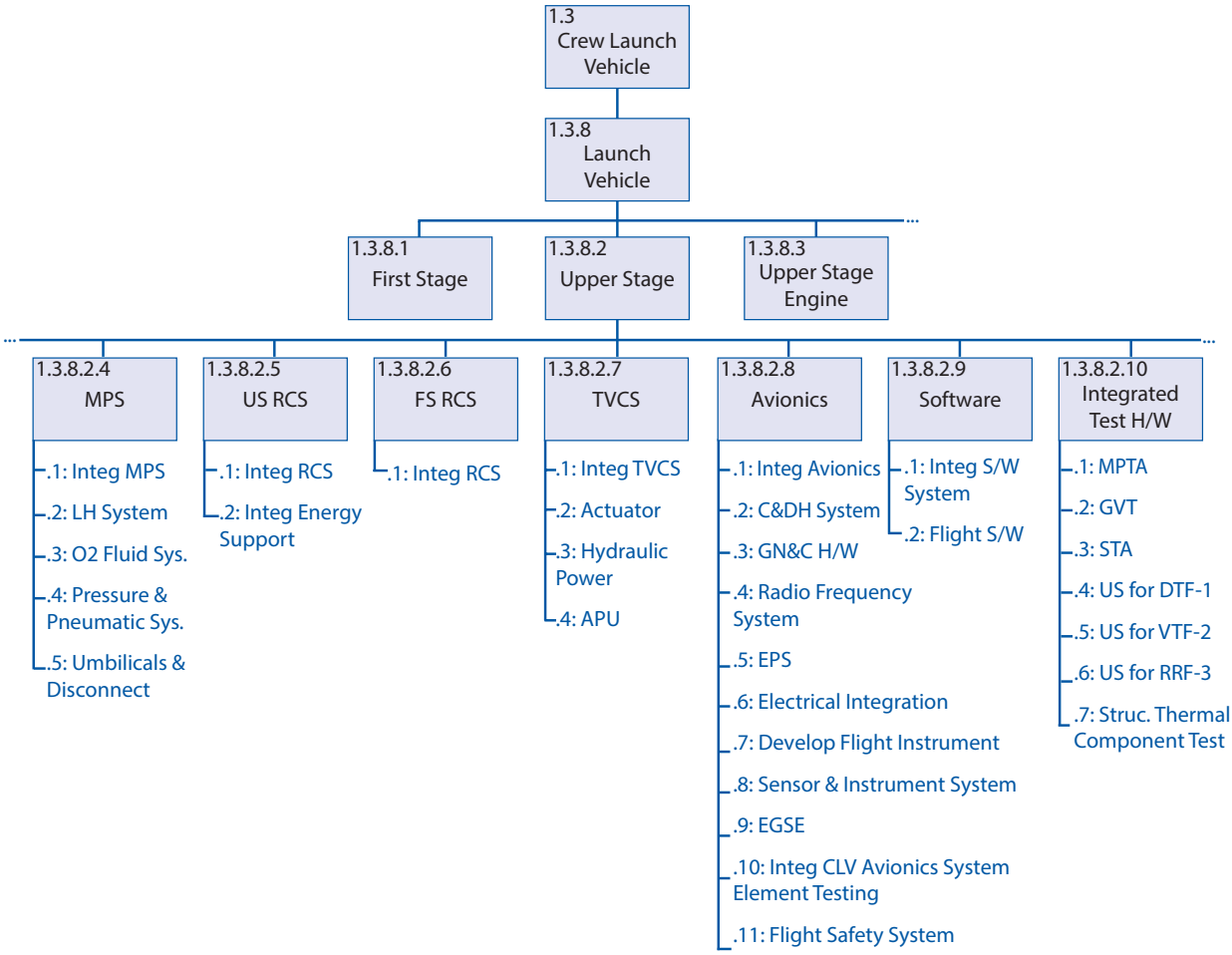


FIGURE G.1-1 PBS Example

technological advancements required. Therefore, at the beginning of Pre-Phase A, the following should be provided:

- Refinement of TRL definitions.
- Definition of AD^2.
- Definition of terms to be used in the assessment process.
- Establishment of meaningful evaluation criteria and metrics that will allow for clear identification of gaps and shortfalls in performance.
- Establishment of the TA team.
- Establishment of an independent TA review team.

G.3 How to Do Technology Assessment

The technology assessment process makes use of basic systems engineering principles and processes. As mentioned previously, it is structured to occur within the framework of the Product Breakdown Structure (PBS) to facilitate incorporation of the results. Using the PBS as a framework has a twofold benefit—it breaks the "problem" down into systems, subsystems, and components that can be more accurately assessed; and it provides the results of the assessment in a format that can be readily used in the generation

of program costs and schedules. It can also be highly beneficial in providing milestones and metrics for progress tracking using EVM. As discussed above, it is a two-step process comprised of (1) the determination of the current technological maturity in terms of TRLs and (2) the determination of the difficulty associated with moving a technology from one TRL to the next through the use of the AD^2.

Conceptual Level Activities

The overall process is iterative, starting at the conceptual level during program Formulation, establishing the initial identification of critical technologies, and establishing the preliminary cost, schedule, and risk mitigation plans. Continuing on into Phase A, the process is used to establish the baseline maturity, the Technology Development Plan, and the associated costs and schedule. The final TA consists only of the TMA and is used to develop the TRAR, which validates that all elements are at the requisite maturity level. (See FIGURE G.3-1.)

Even at the conceptual level, it is important to use the formalism of a PBS to avoid allowing important technologies to slip through the cracks. Because of the preliminary nature of the concept, the systems, subsystems, and components will be defined at a level that will not permit detailed assessments to be made. The process of performing the assessment, however, is the same as that used for subsequent, more detailed steps that occur later in the program/project where systems are defined in greater detail.

Architectural Studies

Once the concept has been formulated and the initial identification of critical technologies made, it is

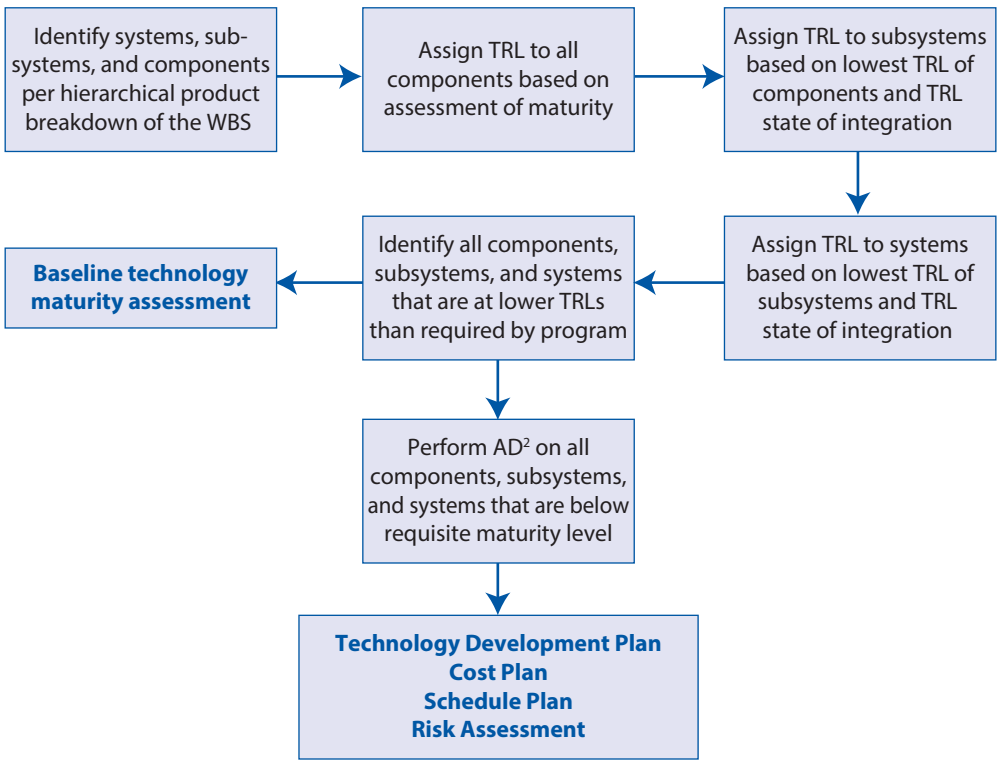


FIGURE G.3-1 Technology Assessment Process

necessary to perform detailed architecture studies with the Technology Assessment Process intimately interwoven. (See FIGURE G.3-2.)

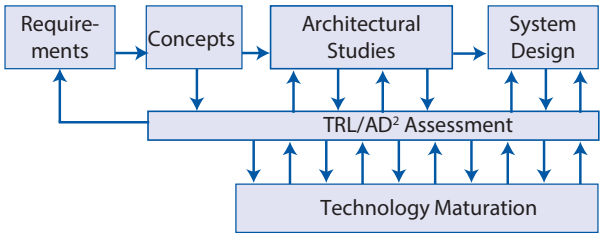


FIGURE G.3-2 Architectural Studies and Technology Development

The purpose of the architecture studies is to refine end-item system design to meet the overall scientific requirements of the mission. It is imperative that there be a continuous relationship between architectural studies and maturing technology advances. The architectural studies should incorporate the results of the technology maturation, planning for alternative paths and identifying new areas required for development as the architecture is refined. Similarly, it is incumbent upon the technology maturation process to identify requirements that are not feasible and development routes that are not fruitful and to transmit that information to the architecture studies in a timely manner. It is also incumbent upon the architecture studies to provide feedback to the technology development process relative to changes in requirements. Particular attention should be given to "heritage" systems in that they are often used in architectures and environments different from those in which they were designed to operate.

G.4 Establishing TRLs

A Technology Readiness Level (TRL) is, at its most basic, a description of the performance history of a given system, subsystem, or component relative to a set of levels first described at NASA HQ in the 1980s. The TRL essentially describes the state of a given technology and provides a baseline from which maturity is gauged and advancement defined. (See FIGURE G.4-1.)

Programs are often undertaken without fully understanding either the maturity of key technologies or what is needed to develop them to the required level. *It is impossible to understand the magnitude and scope of a development program without having a clear understanding of the baseline technological maturity of all elements of the system.* Establishing the TRL is a vital first step on the way to a successful program. A frequent misconception is that in practice, it is too difficult to determine TRLs and that when you do, it is not meaningful. On the contrary, identifying TRLs can be a straightforward systems engineering process of determining what was demonstrated and under what conditions it was demonstrated.

Terminology

At first glance, the TRL descriptions in FIGURE G.4-1 appear to be straightforward. It is in the process of trying to assign levels that problems arise. A primary cause of difficulty is in terminology; e.g., everyone knows what a breadboard is, but not everyone has the same definition. Also, what is a "relevant environment?" What is relevant to one application may or may not be relevant to another. Many of these terms originated in various branches of engineering and had, at the time, very specific meanings to that particular field. They have since become commonly used throughout the engineering field and often acquire differences in meaning from discipline to discipline, some differences subtle, some not so subtle. "Breadboard," for example, comes from electrical engineering where the original use referred to checking out the functional design of an electrical circuit by populating a "breadboard" with components to verify that the design operated as anticipated. Other terms come from mechanical engineering, referring

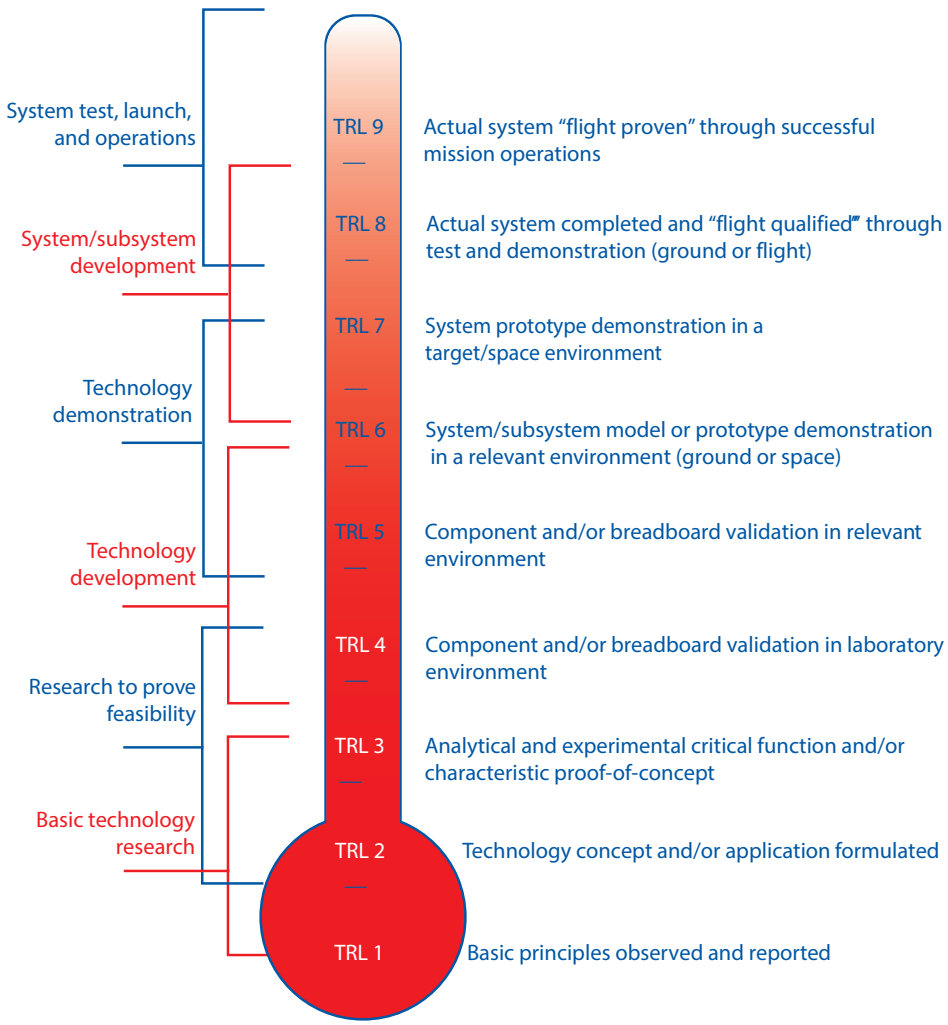


FIGURE G.4-1 Technology Readiness Levels

primarily to units that are subjected to different levels of stress under testing, e.g., qualification, protoflight, and flight units. The first step in developing a uniform TRL assessment (see **FIGURE G.4-2**) is to define the terms used. It is extremely important to develop and use a consistent set of definitions over the course of the program/project.

Judgment Calls

Having established a common set of terminology, it is necessary to proceed to the next step: quantifying "judgment calls" on the basis of past experience. Even with clear definitions, judgment calls will be required when it comes time to assess just how similar a given element is relative to what is needed (i.e., is it close enough to a prototype to be considered a prototype, or is it more like an engineering breadboard?). Describing what has been done in terms of form, fit, and function provides a means of quantifying an element based on its design intent and subsequent performance. The current definitions for software TRLs are contained in NPR 7123.1, NASA Systems Engineering Processes and Requirements.

Assessment Team

A third critical element of any assessment relates to the question of who is in the best position to make judgment calls relative to the status of the technology in question. For this step, it is extremely important to have a well-balanced, experienced assessment team. Team members do not necessarily have to be discipline experts. The primary expertise required for a TRL assessment is that the systems engineer/user understands the current state of the art in applications. User considerations are evaluated by HFE personnel who understand the challenges of technology insertions at various stages of the product life cycle. Having established a set of definitions, defined a process for quantifying judgment calls, and assembled an expert assessment team, the process primarily consists of asking the right questions. The flowchart depicted in FIGURE G.4-2 demonstrates the questions to ask to determine TRL at any level in the assessment.

Heritage Systems

Note the second box particularly refers to heritage systems. If the architecture and the environment have changed, then the TRL drops to TRL 5—at least initially. Additional testing may need to be done for heritage systems for the new use or new environment. If in subsequent analysis the new environment is sufficiently close to the old environment or the new architecture sufficiently close to the old architecture, then the resulting evaluation could be TRL 6 or 7, but the most important thing to realize is that it is no longer at TRL 9. Applying this process at the system level and then proceeding to lower levels of subsystem and component identifies those elements that require development and sets the stage for the subsequent phase, determining the AD^2.

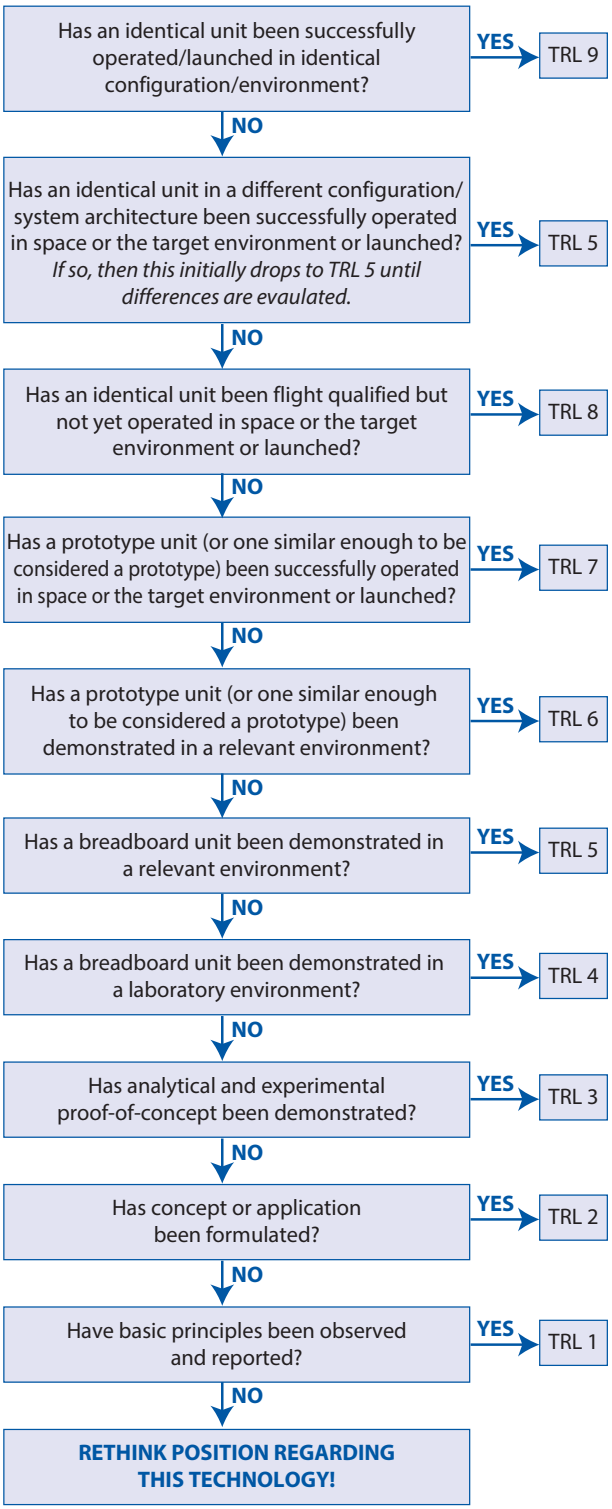


FIGURE G.4-2 TMA Thought Process

Formal Process for Determining TRLs

A method for formalizing this process is shown in FIGURE G.4-3. Here, the process has been set up as a table: the rows identify the systems, subsystems, and components that are under assessment. The columns identify the categories that will be used to determine the TRL; i.e., what units have been built, to what scale, and in what environment have they been tested. Answers to these questions determine the TRL of an item under consideration. The TRL of the system is determined by the lowest TRL present in the system; i.e., a system is at TRL 2 if any single element in the system is at TRL 2. The problem of multiple elements being at low TRLs is dealt with in the AD^2 process. Note that the issue of integration affects the TRL of every system, subsystem, and component. All of the elements can be at a higher TRL, but if they have never been integrated as a unit, the TRL will be lower for the unit. How much lower depends on the complexity of the integration. The assessed complexity depends upon the combined judgment of the engineers. It is important to have a good cross-section of senior people sitting in judgment.

Red = Below TRL 3
Yellow = TRL 3, 4 & 5
Green = TRL 6 and above
White = Unknown
X = Exists

	Concept	Demonstration Units					Environment				Unit Description				Overall TRL
		Breadboard	Brassboard	Developmental Model	Prototype	Flight Qualified	Laboratory Environment	Relevant Environment	Space Environment	Space Launch Operation	Form	Fit	Function	Appropriatae Scale	
1.0 System															
1.1 Subsystem X															
1.1.1 Mechanical Components															
1.1.2 Mechanical Systems															
1.1.3 Electrical Components					X				X		X	X	X		
1.1.4 Electrical Systems															
1.1.5 Control Systems															
1.1.6 Thermal Systems							X				X	X			
1.1.7 Fluid Systems		X													
1.1.8 Optical Systems															
1.1.9 Electro-Optical Systems															
1.1.10 Software Systems															
1.1.11 Mechanisms	X														
1.1.12 Integration															
1.2 Subsystem Y															
1.2.1 Mechanical Components															

FIGURE G.4-3 TRL Assessment Matrix